ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CSE-Cloud Computing)

Semester: II

SUBJECT CODE: PHY107

PHYSICS FOR ENGINEERS –II

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(14 Marks)**

Q. 1. Answer the following questions: **(1×7 = 7 Marks)**

(i) Write the postulates of the special theory of relativity.

(ii) What is the meaning of degeneracy in quantum mechanics?

(iii) Write the Lorentz transformation relations.

(iv) Sketch the one-dimensional infinite square well and barrier potential with mathematical form.

(v) What is the difference between inertial and non-inertial frames of reference?

(vi) What is wave function of first exited state of one-dimensional infinite square well?

(vii) What is Meissner effect?

Q. 2. Answer the following questions: **(1×7 = 7 Marks)**

(i) What is Compton Effect?

(ii) What is the wave-particle duality of Light?

(iii) Define Fermi-energy.

(iv) What is Tunnel effect?

(v) Which type if superconductor shows partial Meissner effect?

(vi) Write the relativistic velocity addition formula.

(vii) By which quantum mechanical process Alpha decay takes place?

SECTION – B

NOTE:Attempt all questions. **(7×2×2 = 28 Marks)**

Q.3.(i)(a) Derive the energy expression and wave function for one dimensional infinite square well.

OR

(b) Write the time-dependent Schrodinger wave equation in one dimension and separate its time-independent part and time-dependent part.

(ii)(a) Explain the process of alpha decay.

OR

(b) Discuss the band theory of solid.

Q.4.(i)(a) What is Josephson Effect? Discuss in brief.

OR

(b) Using band theory of solid, differentiate between conductor, semiconductor and insulator.

(ii)(a) Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$

OR

(b) Compare Type I and Type II superconductor.

SECTION – C

NOTE: Attempt all questions. **(14×2 = 28 Marks)**

Q.5. (a) Explain the construction and working principle of Scintillation counter.

OR

(b) Calculate the divergence of following vectors;

$\vec{A}=x^2\,\hat{i}+3xz^2\,\hat{j}-2xz\,\hat{k}$ $\vec{B}=xy\,\hat{i}+2yz\,\hat{j}+3xz\,\hat{k}$

$\vec{C}=y^2\,\hat{i}+(2xy+z^2\,\hat{j}+2yz\,\hat{k}$

Q.6. (a) Derive the expression of $E_o = m_o c^2$.

OR

(b) Write all Maxwell equation and explain its physical meaning.